



INSTITUTO POLITÉCNICO NACIONAL

UNIDAD PROFESIONAL INTERDISCIPLINARIA DE INGENIERÍA EN TECNOLOGÍAS AVANZADAS

Ingeniería Telemática Bases de Datos Distribuidas Practica 2: Planes de ejecución.

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- **Grupo:** 3TM3
- **Fecha de entrega:** 29 de mayo 2025
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Practica 2 "PLANES DE EJECUCIÓN"

1.- Crear una base de datos con el nombre: practicaPE

2.- Copiar a la base de datos practicaPE las siguientes tablas de la base de datos AdventureWorks

<pre>---/ a) Sales.SalesOrderHeader select * into practicaPE.dbo.SalesOrderHeader from AdventureWorks2019.Sales.SalesOrderHeader; --// select * from SalesOrderHeader --(31,465 rows)</pre>
<pre>---/ b) Sales.SalesOrderHeader select * into practicaPE.dbo.SalesOrderDetail from AdventureWorks2019.Sales.SalesOrderDetail; --// select * from SalesOrderDetail --(121,317 rows)</pre>
<pre>---/ c) Sales.Customer select * into practicaPE.dbo.SalesCustomer from AdventureWorks2019.Sales.Customer; --// select * from SalesCustomer --(19,820 rows)</pre>
<pre>---/ d) Sales.SalesTerritory select * into practicaPE.dbo.SalesTerritory from AdventureWorks2019.Sales.SalesTerritory; --// select * from SalesTerritory --(10 rows)</pre>
<pre>---/ e) Production.Product select * into practicaPE.dbo.ProductionProduct from AdventureWorks2019.Production.Product; --// select * from ProductionProduct --(504 rows)</pre>
<pre>---/ f) Production.ProductCategory select * into practicaPE.dbo.ProductionProductCategory from AdventureWorks2019.Production.ProductCategory; --// select * from ProductionProductCategory --(4 rows)</pre>
<pre>---/ g) Production.ProductSubcategory select * into practicaPE.dbo.ProductionProductSubcategory from AdventureWorks2019.Production.ProductSubcategory;</pre>

```
--// select * from ProductionProductSubcategory --(37
rows)

---/ h) PersonPerson
      select BusinessEntityID, FirstName, LastName
      into practicaPE.dbo.PersonPerson
      from   AdventureWorks2019.Person.Person;   --
(additionalcotactinfo is type with a shema collection)
--// select * from PersonPerson --(19,972 rows)
```

3.- Codificar las siguientes consultas

--a. Listar el producto más vendido de cada una de las categorías registradas en la base de datos.

```
WITH VentasProducto AS (
  SELECT
    pc.Name AS Categoria,
    p.ProductID,
    p.Name AS Producto,
    SUM(od.OrderQty) AS CantidadVendida
  FROM SalesOrderDetail od
  JOIN Product p ON od.ProductID = p.ProductID
  JOIN ProductSubcategory ps ON p.ProductSubcategoryID = ps.ProductSubcategoryID
  JOIN ProductCategory pc ON ps.ProductCategoryID = pc.ProductCategoryID
  GROUP BY pc.Name, p.ProductID, p.Name
),
ProductoMaximo AS (
  SELECT
    Categoria,
    MAX(CantidadVendida) AS MaxCantidad
  FROM VentasProducto
  GROUP BY Categoria
)
SELECT vp.Categoria, vp.Producto, vp.CantidadVendida
FROM VentasProducto vp
JOIN ProductoMaximo pm
  ON vp.Categoria = pm.Categoria AND vp.CantidadVendida = pm.MaxCantidad
ORDER BY vp.Categoria;
```

--b. Listar el nombre de los clientes con más ordenes por cada uno de los territorios registrados en la base de datos.

```
WITH Customer_Orders AS (
  SELECT
    soh.SalesOrderID,
    c.CustomerID,
    soh.TerritoryID,
    p.FirstName,
    p.LastName
```

```

FROM SalesOrderHeader soh
JOIN Customer c ON soh.CustomerID = c.CustomerID
JOIN Person p ON c.PersonID = p.BusinessEntityID
),
OrdenesPorCliente AS (
    SELECT
        TerritoryID,
        CustomerID,
        FirstName,
        LastName,
        COUNT(*) AS Orders
    FROM Customer_Orders
    GROUP BY TerritoryID, CustomerID, FirstName, LastName
),
MaxOrdenesPorTerritorio AS (
    SELECT
        TerritoryID,
        MAX(Orders) AS MaxOrders
    FROM OrdenesPorCliente
    GROUP BY TerritoryID
)
SELECT
    o.TerritoryID,
    o.FirstName,
    o.LastName,
    o.Orders
FROM OrdenesPorCliente o
JOIN MaxOrdenesPorTerritorio m
    ON o.TerritoryID = m.TerritoryID AND o.Orders = m.MaxOrders
ORDER BY o.TerritoryID;

```

--c. Listar los datos generales de las ordenes que tengan al menos los mismos productos de la orden con salesorderid = 43676.

```

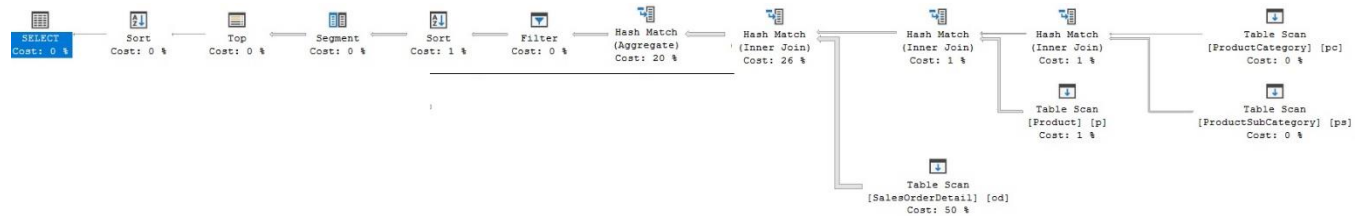
SELECT distinct SalesOrderID, OrderQty, ProductID
FROM SalesOrderDetail AS OD
WHERE NOT EXISTS
    (SELECT *
    FROM ( select ProductID
            from SalesOrderDetail
            where SalesOrderID = 43676) as P
    WHERE NOT EXISTS(
        (select *
          From SalesOrderDetail as OD2
         where OD.salesorderid =OD2.SalesOrderID
          and OD2.ProductID = P.ProductID)));

```

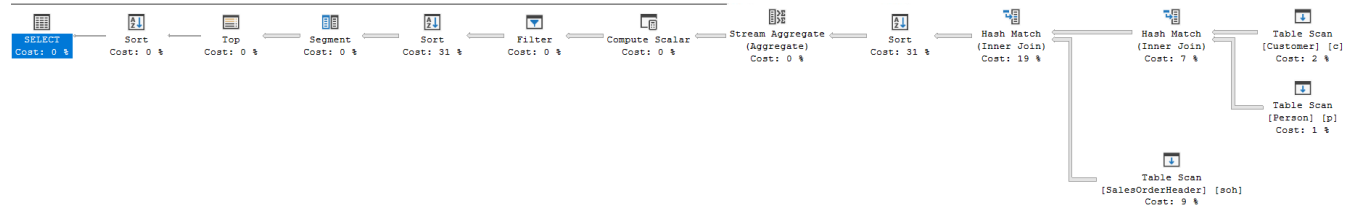
4.- Generar los planes de ejecución de las consultas en la base de datos practicaPE y proponer índices para mejorar el rendimiento de las consultas.

Sin indices:

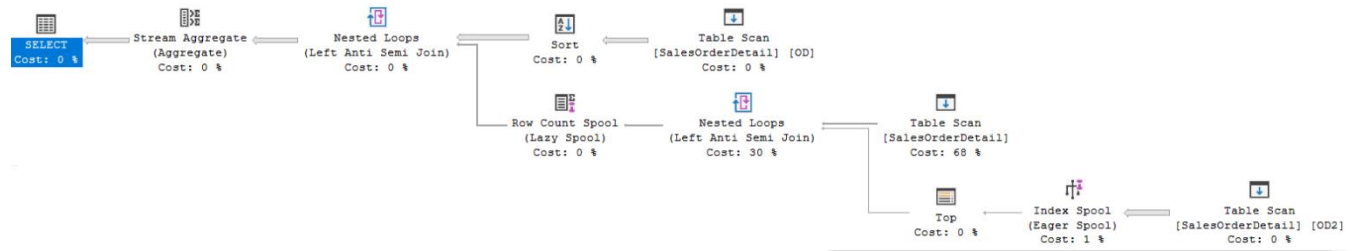
A)



B)

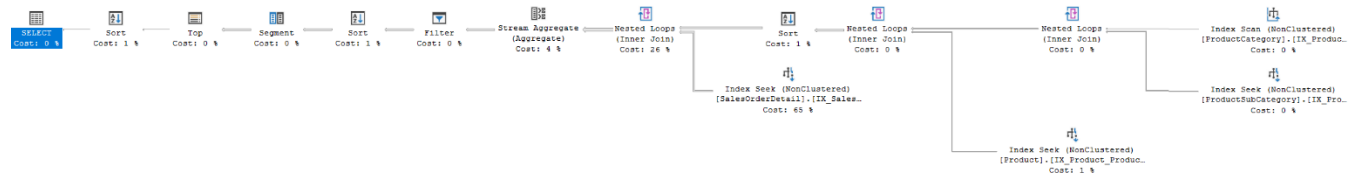


C)



Con índices:

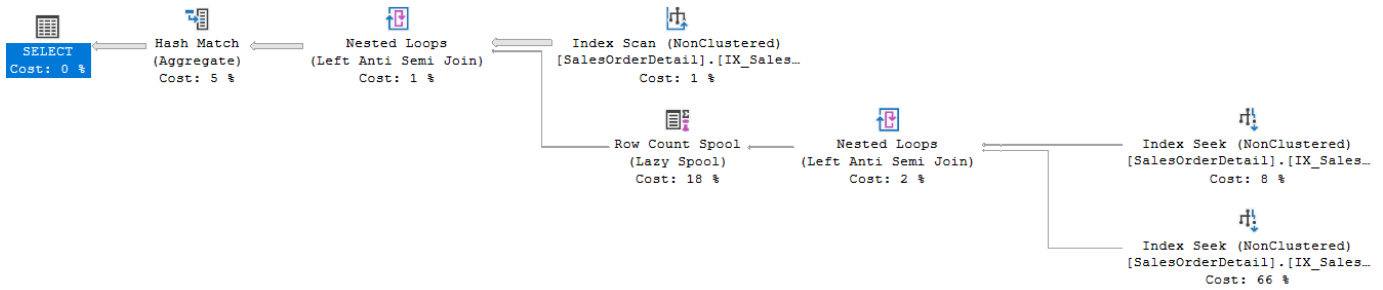
A)



B)

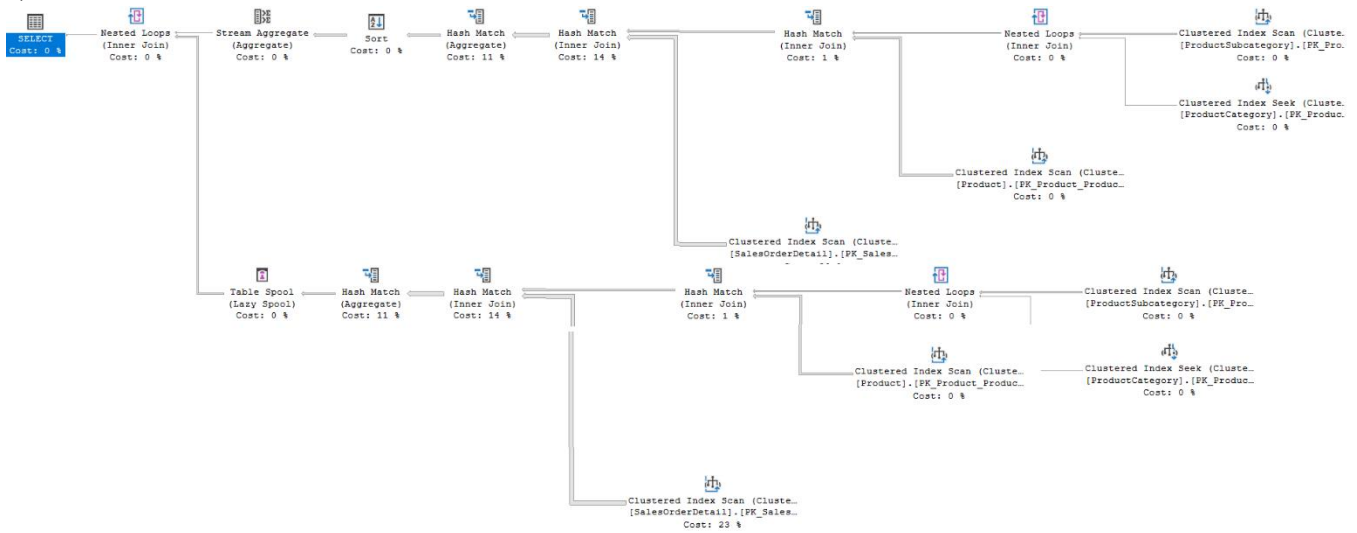


C)



5.- Generar los planes de ejecución de las consultas en la base de datos AdventureWorks y comparar con los planes de ejecución del punto 4.

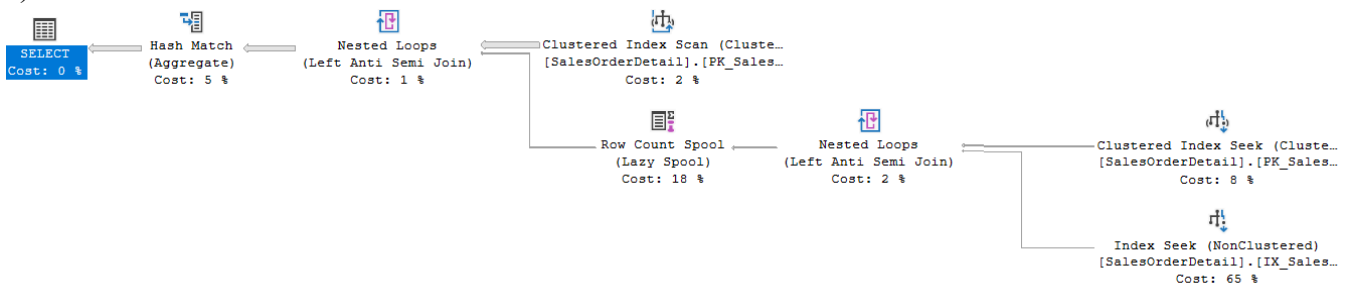
a)



b)



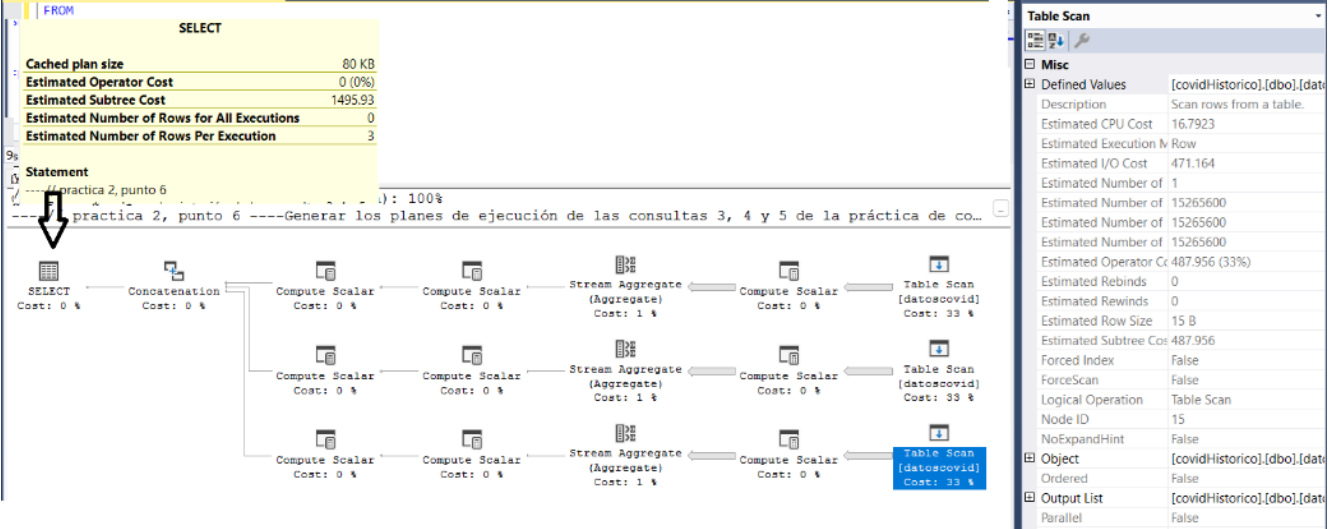
c)



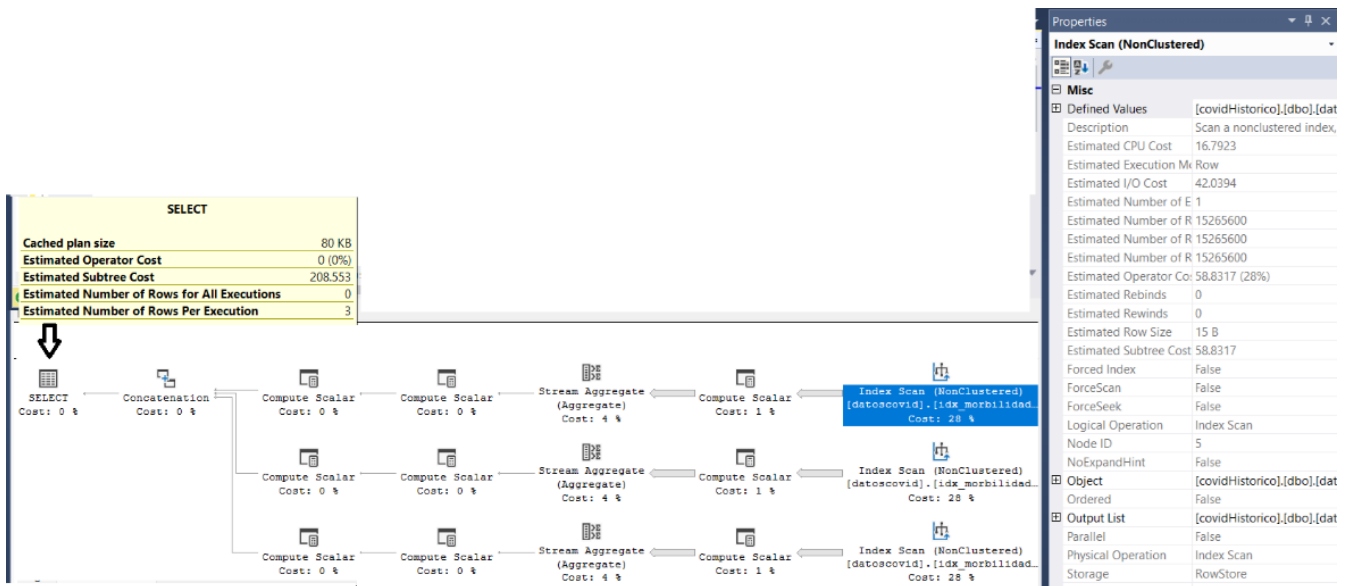
6.- Generar los planes de ejecución de las consultas 3, 4 y 5 de la práctica de consultas en la base de datos Covid y proponer índices para mejorar el rendimiento.

-- Consulta 3; índice propuesto;

```
CREATE INDEX idx_morbididades_confirmados
ON datoscovid (clasificacion_final)
INCLUDE (diabetes, obesidad, hipertension);
```



Con índice;



-- Consulta 4; indice propuesto;

```
CREATE INDEX idx_filtros_con_include
ON datoscovid (CLASIFICACION_FINAL, HIPERTENSION, OBESIDAD, DIABETES,
TABQUISMO)
INCLUDE (Entidad_Res, Municipio_Res);
```

SELECT
Cost: 0 %

Hash Match (Aggregate)
Cost: 7 %

Table Scan [datoscovid]
Cost: 93 %

SELECT

Property	Value
Cached plan size	32 KB
Estimated Operator Cost	0 (0%)
Estimated Subtree Cost	525.43
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	10979.2

Defined Values [covidHistorico].[dbo].[date]

Property	Value
Description	Scan rows from a table.
Estimated CPU Cost	16.7923
Estimated Execution Mode	Row
Estimated I/O Cost	471.164
Estimated Number of Executions	1
Estimated Number of Rows for All Executions	111298
Estimated Number of Rows Per Execution	111298
Estimated Number of Rows to be Read	15265600
Estimated Operator Cost	487.956 (93%)
Estimated Rebinds	0
Estimated Rewinds	0
Estimated Row Size	43 B
Estimated Subtree Cost	487.956
Forced Index	False
ForceScan	False
Logical Operation	Table Scan
Node ID	1
NoExpandHint	False

Object [covidHistorico].[dbo].[date]

Output List [covidHistorico].[dbo].[date]

Parallel False

Physical Operation Table Scan

Predicate [covidHistorico].[dbo].[date]

Storage RowStore

Con índice

SELECT
Cost: 0 %

Hash Match (Aggregate)
Cost: 30 %

Index Scan (NonClustered) [datoscovid].[idx_filtros_con_include]
Cost: 70 %

SELECT

Property	Value
Cached plan size	32 KB
Estimated Operator Cost	0 (0%)
Estimated Subtree Cost	124.504
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	10979.2

Statement

```
select distinct
Entidad_Res, Municipio_Res
from
datoscovid
where
CLASIFICACION_FINAL not in(1,2,3) and HIPERTENSION=1
and OBESIDAD=1 and diabetes=1 and TABQUISMO=1
```

Properties

Index Scan (NonClustered)

Misc

Defined Values [covidHistorico].[dbo].[datoscovid].[ENTIDAD_RES, [o]

Property	Value
Description	Scan a nonclustered index, entirely or only a range.
Estimated CPU Cost	16.7923
Estimated Execution Mode	Row
Estimated I/O Cost	70.2379
Estimated Number of Executions	1
Estimated Number of Rows for All Executions	111298
Estimated Number of Rows Per Execution	111298
Estimated Number of Rows to be Read	15265600
Estimated Operator Cost	87.0303 (70%)
Estimated Rebinds	0
Estimated Rewinds	0
Estimated Row Size	43 B
Estimated Subtree Cost	87.0303
Forced Index	False
ForceScan	False
ForceSeek	False
Logical Operation	Index Scan
Node ID	1
NoExpandHint	False

Object [covidHistorico].[dbo].[datoscovid].[idx_filtros_con_in

Ordered False

Output List [covidHistorico].[dbo].[datoscovid].[ENTIDAD_RES, [o]

Parallel False

Physical Operation Index Scan

Predicate [covidHistorico].[dbo].[datoscovid].[CLASIFICACION_

Storage RowStore

Table Cardinality 15265600

Defined Values

Contains a comma-separated list of values introduced by this operator, which may be computed ex

-- Consulta 5; indice propuesto;

```
CREATE INDEX idx_casos_recuperados_neumonia_inc
ON datoscovid (CLASIFICACION_FINAL, FECHA_DEF, NEUMONIA)
INCLUDE (ENTIDAD_RES);
```

Execution Plan Summary:

- Table Scan [datoscovid]**: Cost: 94. Properties: Description: Scan rows from a table. Estimated CPU Cost: 16.7923. Estimated Execution Mode: Row. Estimated I/O Cost: 471.164. Estimated Number of Rows: 442586. Estimated Number of Rows for All Executions: 442586. Estimated Number of Rows to be Read: 15265600. Estimated Operator Cost: 487.956 (94%). Estimated Rewinds: 0. Estimated Row Size: 45 B. Estimated Subtree Cost: 487.956. Forced Index: False. ForceScan: False. Logical Operation: Table Scan. Node ID: 3. NoExpandHint: False. Object: [covidHistorico].[dbo].[datoscovid]. Ordered: False.
- Hash Match (Aggregate)**: Cost: 6.
- Compute Scalar**: Cost: 0.
- Sort (Top N Sort)**: Cost: 0.
- SELECT**: Cost: 0. Properties: Cached plan size: 40 KB. Estimated Operator Cost: 0 (0%). Estimated Subtree Cost: 519.641. Estimated Number of Rows for All Executions: 0. Estimated Number of Rows Per Execution: 3.

Con índices

Execution Plan Summary:

- Index Seek (NonClustered) [datoscovid].[idx_casos_recu...]**: Cost: 47. Properties: Description: Scan a particular range of rows from a nonclustered index. Estimated CPU Cost: 0.485339. Estimated Execution Mode: Row. Estimated I/O Cost: 2.11053. Estimated Number of Executions: 1. Estimated Number of Rows for All Executions: 441075. Estimated Number of Rows Per Execution: 441075. Estimated Number of Rows to be Read: 441075. Estimated Operator Cost: 2.59587 (47%). Estimated Rewinds: 0. Estimated Row Size: 15 B. Estimated Subtree Cost: 2.59587. Forced Index: False. ForceScan: False.
- Hash Match (Aggregate)**: Cost: 53.
- Compute Scalar**: Cost: 0.
- Sort (Top N Sort)**: Cost: 0.
- SELECT**: Cost: 0. Properties: Cached plan size: 48 KB. Estimated Operator Cost: 0 (0%). Estimated Subtree Cost: 5.57172. Estimated Number of Rows for All Executions: 0. Estimated Number of Rows Per Execution: 3.

--// 7.- Comparar los planes de ejecución del punto 6 con los planes de ejecución de otro equipo.

--// 8.- Conclusiones: